

1st SIT EXAMINATION QUESTION PAPER:

Year Long 2023 - 2024

Module Code:	CC6001NI
Module Title:	Advanced Database Systems Development
Module Leader:	Mr. Lekhnath Katuwal (Islington College)

Date:	
Day / Evening:	Day
Start Time:	
Duration:	2 Hours

Exam Type:	Unseen, Closed
Materials supplied:	None
Materials permitted:	Writing equipment only
Warning:	Candidates are warned that possession of unauthorised materials in an examination is a serious assessment offence.

Instructions to candidates:	Please Note: Inclusive of this cover page, this examination paper consists of 5 pages. The student must complete four (4) out of five (5) questions. This examination accounts for 40% of your total module grades. You are to return this examination paper, BEFORE you leave the examination room. DO NOT TURN PAGE OVER UNTIL INSTRUCTED
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Attempt Any Four Questions.

Question 1

1.a) Define anomalies and its type with suitable example.

[2+1+3 = 6 Marks]

1.b) Write short notes on following topics

[2x5 = 10 Marks]

- i) Full Functional Dependency
- ii) Rollback
- iii) Commit
- iv) Cross Join
- v) Any two techniques for Data Mining

1.c) Normalize the given table up to 3NF indicating all the dependency and Keys (primary and foreign) clearly.

Note: State any assumption that you have made during normalization.

[1+ 2+ 3+ 3 = 9 Marks]

Shipment_no	Date	Origin	Destination	Item_no	Description	Weight	Supplier_ID	Supplier_Name	Qty
39	1/10/2010	Boston	Brazil	3223	BM Concrete	500	S1	ABC Company	100
				3297	MB Steel	90	S2	IG	2000
40	1/11/2010	Boston	Atlanta	3568	Plywood	100	S3	XYZ Company	200
				3297	MB Steel	90	S2	IG	1500
41	1/12/2010	Chicago	Dallas	3223	BM Concrete	500	S1	ABC Company	150
				3672	Concrete Nail	150	S4	Happy Builders	200

Question 2

2. a) In the course of transaction processing, the system takes checkpoints at specified intervals. Assume that sometime after a particular checkpoint was taken a system failure occurred. Consider the following transactions in the scenario:

T1 started after the checkpoint and did not complete when system failed
T2 started and completed before checkpoint was taken
T3 started before checkpoint and did not complete when system failed
T4 started after the checkpoint and was completed before the system failed
T4 started before the checkpoint and was completed before system failed

(i) Draw a diagram to illustrate the above set of transactions.

[5 Marks]

(ii) For each of the above transactions, indicate which must be redone, which must be undone and which is unaffected by the failure.

[5 Marks]

2. b) Define Transaction and its ACID properties.

[2 + 4 = 6 Marks]

2. c) Explain the process to recover from both transaction failure and System failure in case of Deferred update and immediate update respectively.

[4.5 + 4.5 = 9 Marks]

Question 3

3. a) Define Deadlock with suitable example.

[2+2 = 4 Marks]

3. b) Complete the following transaction with suitable application of Locks.

[4 Marks]

[Note: If deadlock encountered, no need to resolve]

Time	T _A	T _B
t1	begin	
t2	read (balance)	
t3	balance=balance+200	
t4	write (balance)	begin
t5		read (balance)
t6	rollback	balance=balance-300
t7		write (balance)
t8		commit

3. c) The following list represents the sequence of events in an interleaved execution of a set of transactions T1, T2, ... T24 in a concurrency system based on locking, where A, B, ...H are data items. Assume that FETCH A acquires an S lock on A, UPDATE A promotes that lock to an X lock, and all locks are held to the next synch point.

time	transaction	event
t1	T1	FETCH A
t2	T1	FETCH B
t3	T2	FETCH C
t4	T2	FETCH B
t5	T1	UPDATE B
t6	T2	UPDATE C
t7	T2	COMMIT
t8	T3	FETCH B
t9	T4	FETCH A
t10	T4	FETCH D
t11	T5	FETCH G
t12	T5	UPDATE G
t13	T5	FETCH B
t14	T1	FETCH E
t15	T4	UPDATE D
t16	T1	UPDATE E
t17	T1	FETCH D
t18	T4	FETCH F
t19	T6	FETCH F
t20	T4	UPDATE F
t21	T6	FETCH B
t22	T7	FETCH H
t23	T8	FETCH F
t24	T8	FETCH B
t25	T9	FETCH A
t26	T9	COMMIT
t27		

- i) Provide a Wait-for-Graph to illustrate whether there is any deadlock at time *t27*. You are advised to provide full and appropriate workings.

[12 Marks]

- ii) Discuss how the system could recover if it were deadlocked at time *t27*, and justify your choice of victim transaction(s).

[5 Marks]

Question 4

*Please, refer to the **APPENDIX** section for the schema.*

4.a) List the sequence in order to create a table from a given schema.

[3 Marks]

4.b) Write a SQL statement for the following: -

i) To create 'Order_product' table from the above schema with required keys.

[3 Marks]

ii) Find the details of the most ordered product in terms of quantity.

[5 Marks]

iii) Find the name and email of those person who are both customers as well as staff.

[3 Marks]

iv) Find the name of staff hired before 7 years and salary more than 100000.

[3 Marks]

4.c) Write a relational algebra for the following: -

i) Find the name of customer from 'Paris' city.

[3 Marks]

ii) Find the name and email of those person who are both customer as well as staff and gender are 'Female' and name end with 'a'.

[5 marks]

Question 5

5.a) Explain the difference between Database and Database Warehouse.

[2 Marks]

5.b) There are several options available for distributing data in a distributed database system. Describe two options for data fragmentation (horizontal fragmentation and vertical fragmentation). For each option, give an example to illustrate your answer.

[8 Marks]

5.c) Consider the schema from APPENDIX section:

i. Create two users as 'QA' and 'Admin'

[2 Marks]

ii. Grant all necessary privileges for user 'QA' to read data from Person, customer, product, and order.

[3 Marks]

iii. Grant all CRUD privileges for user 'Admin' to Person, customer, address, product, order.

[4 Marks]

5.d) From above question (**Q. 5.c**), create an Access Matrix in tabular form.

[6 Marks]

Appendix:

Consider the following schema for Oracle DBMS:

Person (person_id, name, dob, gender, contact_no, email, address_id*)

Staff (staff_id*, hire_date, salary, position)

customer (customer_id*, membership_type)

address (address_id, country, state, city, street)

product (product_id, name, description, category, unit_cost)

Order (order_id, order_date, total_cost, customer_id*)

order_product (order_id*, product_id*, quantity)

Time	T _A	T _B
t1	begin	
t2	read (balance)	
t3	balance=balance+200	
t4	write (balance)	begin
t5		read (balance)
t6	rollback	balance=balance-300
t7		write (balance)
t8		commit

Time	Ta	Tb
T1	begin	
T2	Request for S lock and granted Read (balance)	
T3	balance= balance+ 200	
T4	Request for X lock and Granted Write (balance)	begin
T5	...	Request for S lock but denied Wait
T6	Rollback [End of tranx and release lock on balance]	S lock granted read (balance)
T7		balance = balance - 300
T8		Request for X lock and granted write (balance)
t9		Commit [End of tranx and release of Lock on balance]